Piston Sensing with a Dispersed Photonic Lantern

From how a lantern works to a 7- vs 19-port study for ANDES

(work in progress – placeholders marked in red)

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TBD

Outline

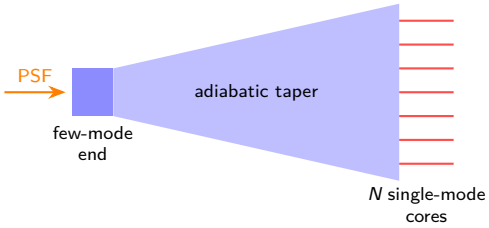
1. What a photonic lantern is and how it works
2. Reconstructors at a single wavelength
3. Case study: ANDES petal pistons with a dispersed lantern
4. Piston sensing with a dispersed lantern: comparison of approaches
5. Conclusions

The problem: limits of pupil-plane sensors

- Shack-Hartmann and pyramid sensors work in or near a pupil plane and measure local gradients of the wavefront phase: large dynamic range, well understood.
- Three problems dominate once the atmosphere is already well corrected:
  1. **Non-common-path aberrations**: invisible to a separate sensing arm.
  2. **Petal / island modes**: piston steps between pupil fragments have (almost) zero gradient where the light is; finite spiders give only a weak edge-diffraction signal.
  3. **Photon loss**: light diverted to the sensor is lost to science.
- A **focal-plane** sensor sees all three – if it can measure phase, not just intensity.

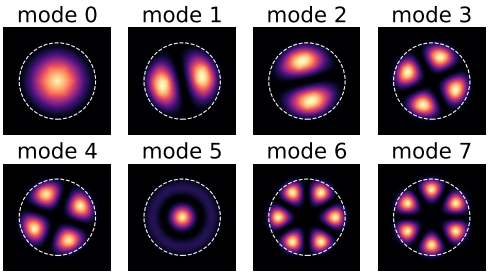
The photonic lantern: a focal-plane wavefront sensor

- A slowly tapered fibre device: **one few-mode end** $\leftrightarrow$ N **single-mode cores**.
- Placed at the **science focus**: no non-common path.
- Its N guided modes **interfere with fixed, device-specific phases** on the way to the cores $\Rightarrow$ the core intensities encode the focal-plane *phase*, including the sign of even aberrations.
- Outputs are single-mode fibres: the same photons can feed a spectrograph.
- **Price**: low order ($\lesssim N$ modes), modest range, device-specific response that must be calibrated or simulated.



Guided (bound) modes and the effective index

- A **mode** is a transverse field pattern that propagates without changing its shape: $E(x, y, z) = \psi(x, y)\, e^{i\beta z}$, with $\nabla_\perp^2 \psi + k_0^2 n^2(x, y)\, \psi = \beta^2 \psi$, $k_0 = 2\pi/\lambda$.
- A **bound (guided) mode** keeps its power in and around the core: its field decays exponentially in the cladding, so it travels along the fibre without radiating. Patterns that are not bound leak their power sideways (radiation modes) within a short distance.
- **Effective index** $n_{\mathrm{eff}} = \beta/k_0$: the mode's phase advances along z as for a plane wave in a uniform medium of index n_{eff}. It is a weighted average of the indices the mode sees: bound modes have $n_2 < n_{\mathrm{eff}} < n_1$; well-confined (low-order) modes are close to n_1, modes near cut-off close to n_2.



The first 8 bound modes at the multimode end of the 19-port lantern, in decreasing n_{eff} (cbeam simulation, 800 nm).

Numerical aperture and the number of bound modes

- Core index n_1, cladding index $n_2 < n_1$. A ray is trapped by total internal reflection if it enters within the acceptance angle $\theta_{\max}$:

$$\mathrm{NA} = n_0 \sin\theta_{\max} = \sqrt{n_1^2 - n_2^2}$$

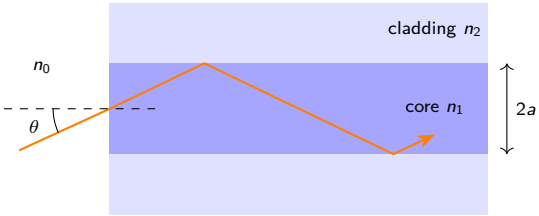
($n_0 = 1$ outside the fibre).

- The number of bound modes is set by the normalised frequency

$$V = \frac{2\pi a}{\lambda}\,\mathrm{NA}, \qquad N_{\mathrm{modes}} \approx V^2/4$$

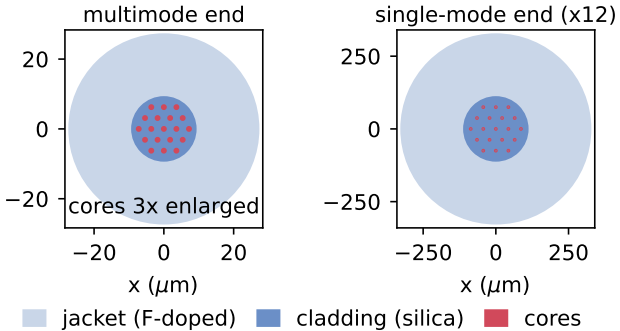
(a: core radius). Single-mode if $V < 2.405$.

- A lantern needs $\approx N$ bound modes at its multimode end. There the role of n_1 is played by the cladding and that of n_2 by the low-index jacket.



rays entering within $\theta \le \theta_{\max}$ are guided

A concrete lantern: the 19-port, 800 nm reference design

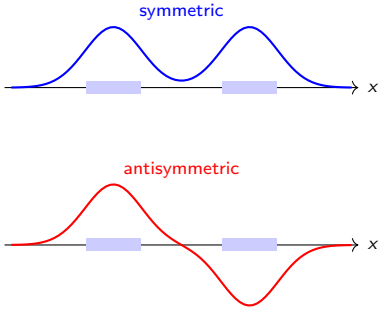


- Cladding (silica, $n = 1.444$) inside a low-index jacket ($\Delta n = -5.5 \times 10^{-3}$): at the waist the cladding *is* the multimode core ($V = 8.9$, $\sim$23 bound modes).
- 19 cores ($\Delta n = +8.8 \times 10^{-3}$), tapered $\times$12 over 100 mm.

cbeam (J. Lin) – batch_pipeline preset p119_800

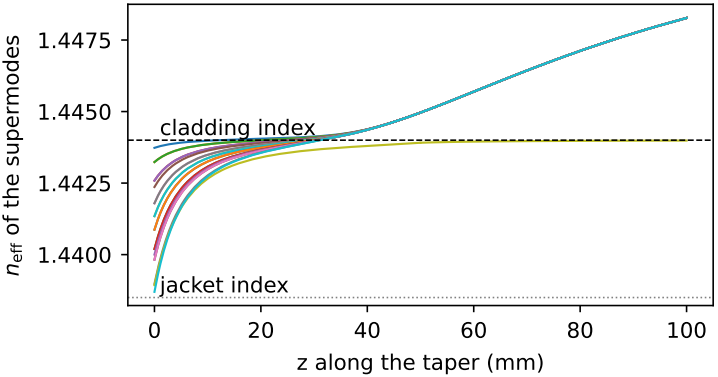
Supermodes

- A **supermode** is a bound mode of the *whole* cross-section – all cores, cladding and jacket together – not of a single core.
- Two identical cores close together: their individual modes overlap and couple; the bound modes of the pair are the symmetric and antisymmetric combinations, with slightly different n_{eff}. With N cores: N supermodes.
- Light launched into one core beats between the supermodes: this is how power moves from core to core (the directional coupler).
- In a lantern the supermodes change along z:
  - waist: cores too small to guide $\Rightarrow$ supermodes $=$ LP modes of the cladding guide;
  - single-mode end: cores far apart $\Rightarrow$ N degenerate supermodes $\approx$ the individual core modes.



the two supermodes of two coupled cores (field)

Inside the taper: from LP modes to core modes



- Multimode end ($z = 0$): cores too small to guide; the supermodes are the LP modes of the cladding guide, well separated in n_{eff}.
- Along the taper the cores grow and capture the light; the supermodes merge.
- Single-mode end: 19 **degenerate** core modes (+ one cladding mode, lost).
- Adiabatic: each supermode follows its own curve, **no power exchange** – but each picks up its own phase $\int \beta \, dz$.

How we simulate it (1): characterizing the lantern

cbeam (J. Lin) solves the scalar, paraxial problem with **coupled-mode theory**: the field is written on the local supermodes $\xi_n(x, y; z)$ of the cross-section at each z,

$$u(x, y, z) = \sum_{n=1}^{N} a_n(z)\, \xi_n(x, y; z).$$

Characterization (once per lantern and wavelength), along an adaptive grid of z values:

1. **Eigen-solve** the cross-section at z with finite elements (mesh scaled by the local taper factor): $\nabla_\perp^2 \xi_n + k_0^2 n^2(x, y; z)\, \xi_n = \beta_n^2 \xi_n \Rightarrow \beta_n(z)$, $\xi_n(z)$ for the N tracked modes.
2. **Track** the modes from one z to the next: match them by overlap, fix their sign, align the bases of degenerate pairs; the step shrinks where the modes change fast.
3. **Couplings** $\kappa_{mn}(z) = \langle \xi_m | \partial_z \xi_n \rangle$ from the z-derivative of the spline-interpolated modes (corrected for the moving mesh); κ is antisymmetric.

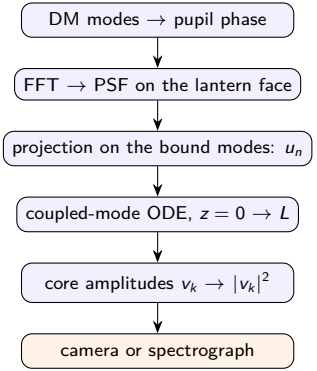
Cost: $\sim$1–2 min per wavelength (7 ports) to $\sim$13 min (19 ports); cached and reused.

How we simulate it (2): propagating fields

- Substituting the expansion in the paraxial equation and projecting on ξ_m gives a **linear ODE for the N amplitudes**. Removing each mode's own fast phase, $a_n = c_n\, e^{i\Theta_n}$ with $\Theta_n(z) = \int_0^z \beta_n\, dz'$:

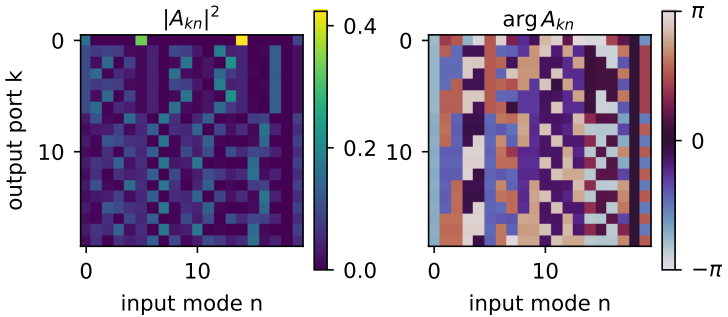
$$\frac{dc_m}{dz} = -\sum_n \kappa_{mn}(z)\, e^{i[\Theta_n(z)-\Theta_m(z)]}\, c_n$$

- Integrated from $z = 0$ to L with an adaptive Runge-Kutta solver (tolerance 10^{-12}), for thousands of input fields at once (one batched call).
- Adiabatic limit ($|\kappa_{mn}| \ll |\beta_n - \beta_m|$): the c_n stay constant – each supermode keeps its power and only accumulates its phase $\Theta_n(L)$.
- At $z = L$: project the supermodes on the individual core modes $\Rightarrow$ the core amplitudes v_k.



50 000 fields × 41 wavelengths: ∼1.5 h on 50 cores within half of the server's memory

The lantern as a matrix: $\mathbf{v} = A\,\mathbf{u}$



- Column n: launch multimode mode n alone, read the 19 cores.
- Each input mode illuminates many cores; the **phases** of A are what make the lantern a wavefront sensor.
- Unitary on the guided subspace (lossless, adiabatic lantern).
- Depends on wavelength, temperature, fabrication: measured or simulated, per device.

The transfer matrix in detail

- Definition: $A_{kn} = \langle \phi_k \,|\, \mathcal{U}(L)\, \psi_n \rangle$ – launch bound mode ψ_n of the multimode end, propagate ($\mathcal{U}$: the coupled-mode solution), project on the mode ϕ_k of output core k. Computed by propagating each unit vector $\mathbf{e}_n$.
- Size: 19 cores $\times$ 20 tracked modes. Mode 18 does not reach any core (it ends as cladding light and is dropped): its column is empty.
- **Unitary** on the 19 guided modes to within 0.7% here: $\sum_k |A_{kn}|^2 = 0.993$–$0.997$ (the rest is numerical; real losses would show as smaller column norms).
- **Phases**: a phase per row is unobservable (we detect $|v_k|^2$); the column phases follow the sign convention of the modes used to compute u_n and must be consistent with it.
- **Structure**: the central core is fed mostly by the radially symmetric modes 14 and 5 (42% and 33% of $|A_{0n}|^2$); degenerate partner modes give pairs of similar columns.
- Depends on wavelength (one A per wavelength: 41 here), temperature and fabrication: every real lantern must be calibrated.
- With the projection written in the pupil, the whole sensor is one matrix: $\mathbf{p} = |AM\mathbf{f}|^2$, $\mathbf{f}$ = pupil field, M = bound modes carried back to the pupil.

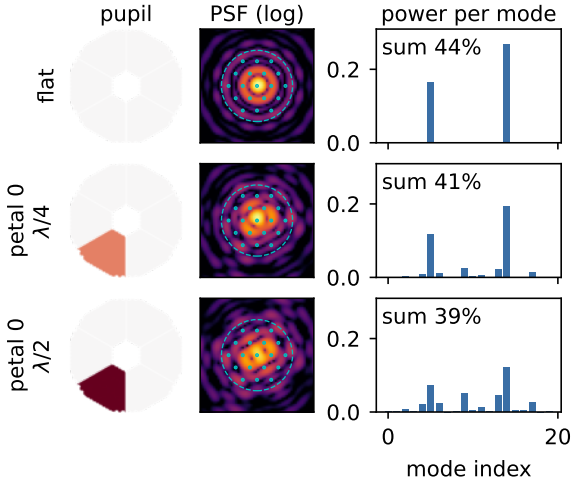
Forward model: each output is an N-beam interferometer

1. Pupil field $P e^{i\varphi} \to$ focal field $E = \mathcal{F}[P e^{i\varphi}]$.
2. Projection on the guided modes: $u_n = \langle \psi_n | E \rangle$.
3. Lantern: $\mathbf{v} = A\mathbf{u}$; detection $p_k = |v_k|^2$.

$$p_k = \sum_n |A_{kn}|^2 |u_n|^2 + 2 \sum_{n<n'} \mathrm{Re}[A_{kn} A_{kn'}^* u_n u_{n'}^*]$$

- The interference terms carry the phase information; the fixed phase offsets $\arg A_{kn} - \arg A_{kn'}$ let a standard lantern see **even** aberrations with sign (a camera, A real, or a mode-selective lantern, A diagonal, cannot).
- Only N intensities per wavelength, normalised: $N-1$ independent numbers. **Keep this in mind for Part 4.**

Example: one petal piston, seen by the lantern



- Left to right: pupil (one petal pushed), PSF on the lantern face (dashed: multimode core; dots: cores), power in each guided mode.
- A petal piston reshapes the PSF and spreads light over many modes; the coupled fraction drops only slowly (44% $\rightarrow$ 39%).
- The first mode (LP_{01}) gets almost no light, even for a flat wavefront – next slide.
- The coupled fraction ($\sim$44%) reflects the PSF size choice of this design (see appendix).

Why does the first mode carry (almost) no light?

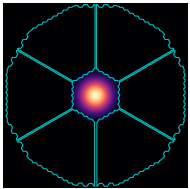
- The coupling is an overlap, and (Parseval) it can be computed in the pupil as well:

$$u_n = \langle \psi_n | E \rangle = \langle \Psi_n \,|\, P\, e^{i\varphi} \rangle,$$

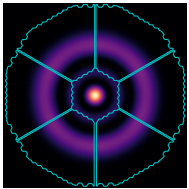
  Ψ_n = mode n carried back to the pupil plane.

- LP_{01} is broad on the lantern face (1/e radius ≈ 6.4 μm vs. Airy first null 2.7 μm) $\Rightarrow$ its pupil image is a small central spot (1/e radius ≈ 0.22 of the pupil radius) – inside the **central obstruction** (0.24–0.28): only 5% of it falls on the open pupil.
- LP_{02}, LP_{03} have rings on the face $\Rightarrow$ annular pupil images that reach the illuminated pupil (80–86%): they take the light.
- A consequence of the small PSF of this design; a larger PSF would couple to LP_{01}.

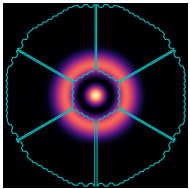
mode 0
5% on the pupil



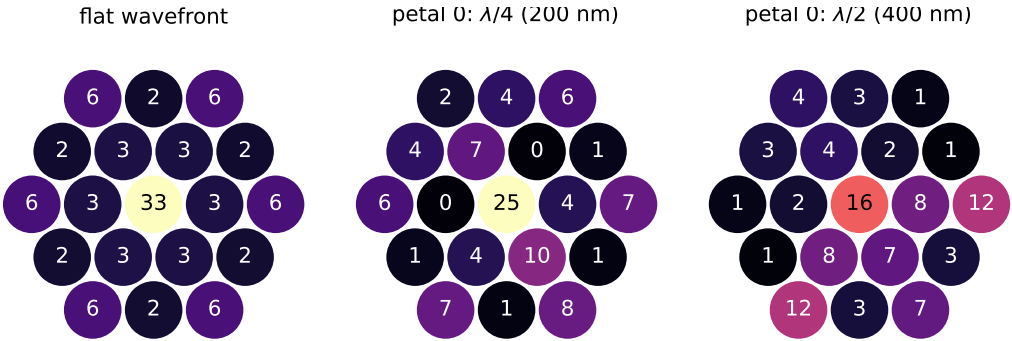
mode 14
86% on the pupil



mode 5
80% on the pupil



…and what comes out of the 19 cores



- Same light, redistributed over the single-mode cores by the interference of the guided modes: the 19 numbers are the sensor signal.
- $\lambda/2$ of piston (right) is *not* the flat case: at one wavelength a piston is seen modulo λ, and the response is periodic.

The inverse problem

- Forward: pistons $\mathbf{p} \in \mathbb{R}^5 \to$ signals $\mathbf{s} = f(\mathbf{p}) \in \mathbb{R}^N$ (smooth, nonlinear, periodic with period λ in each piston).
- Reconstructor: $\hat{\mathbf{p}} = g(\mathbf{s}) \approx f^{-1}(\mathbf{s})$, learned from simulated (or calibration) pairs $(\mathbf{p}, \mathbf{s})$.
- At one wavelength the pistons must stay inside the **capture range** $\pm\lambda/2$; we use all of it: $\pm$400 nm at 800 nm, $\pm$600 nm at 1200 nm. At its edges $+\lambda/2$ and $-\lambda/2$ are the same piston: some cases are ambiguous for any reconstructor.
- Two known degeneracies:
  - **periodicity**: p and $p + \lambda$ give the same signals;
  - **twin image**: $\varphi(x)$ and $-\varphi(-x)$ give the same focal *intensity* – broken by the lantern's mode mixing (not by a camera).
- Evaluation: RMSE per piston on held-out samples (8000 train / 2000 validation, noiseless simulation unless stated).

A family of reconstructors

Approach	**Idea**	**Trade-off**
Linear (interaction matrix)	push-pull calibration, $\hat{\mathbf{p}} = R(\mathbf{s} - \mathbf{s}_0)$	simple, small range
Dense polynomial ridge	all monomials of degree $\leq d$	explainable; size $\sim N^d$
Sparse polynomial (OMP)	keep the best K monomials per piston	explainable, tunable size
Closed-form forward model	exact $\lvert\sum\rvert^2$ structure, Gauss-Newton inversion	physical; needs a good start
Symbolic regression	search for compact formulas (PySR)	readable; slow, approximate
Neural network (MLP)	learn g directly	accurate; opaque, data hungry

- All trained on the same simulated datasets; "degrees of freedom per piston" used as a common size measure.

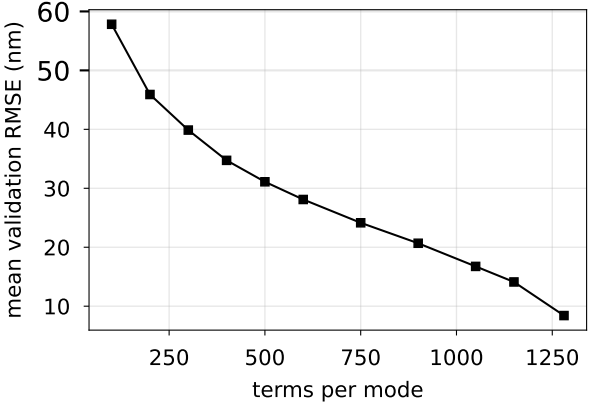
Linear reconstruction and calibration

- Linearise: $\mathbf{p} \approx R(\mathbf{s} - \mathbf{s}_0)$; $R =$ (regularised) pseudo-inverse of the interaction matrix B.
- B by push-pull on the DM: $B_{\cdot m} \approx [\mathbf{s}(+\delta) - \mathbf{s}(-\delta)]/2\delta$.
- Rank $\leq N - 1$ (normalised signals).
- The reference $\mathbf{s}_0$ defines the convergence point.
- Linear range $\lesssim 1$ rad rms.

19-port, 800 nm [old ranges]

Ridge-regularised linear reconstructor: **103 nm** rms (no skill: 173 nm; $\pm$300 nm). The response over $\pm\lambda/2$ is far from linear $\Rightarrow$ nonlinear reconstructors are needed.

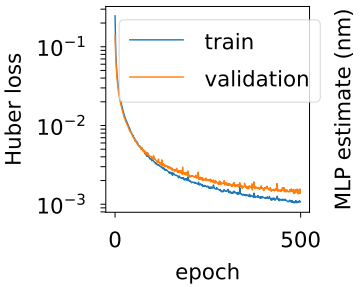
Polynomial reconstructors: how many terms are needed?

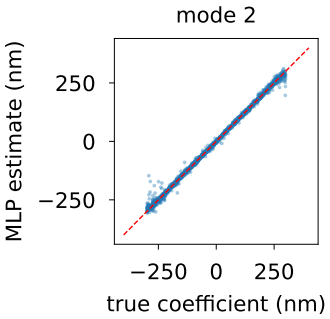


- Orthogonal Matching Pursuit keeps at most K of the 1539 degree-≤ 3 monomials of the 19 signals, per piston.
- The error falls smoothly with K and **saturates at** $K = 1281$: every piston keeps its natural support; larger budgets reproduce the same fit (8.41 nm).
- $K = 1282$ (1281 terms + intercept) is the chosen budget.
- Dense quadratic: 56.5 nm; cubic is needed.

[old ranges] numbers and plots from the first campaign (± 300 / ± 500 nm single, ± 5 μm spectral); the rerun at $\pm\lambda/2$ and $\pm 8\lambda$ is running and replaces them.

Neural network (MLP) reconstructor

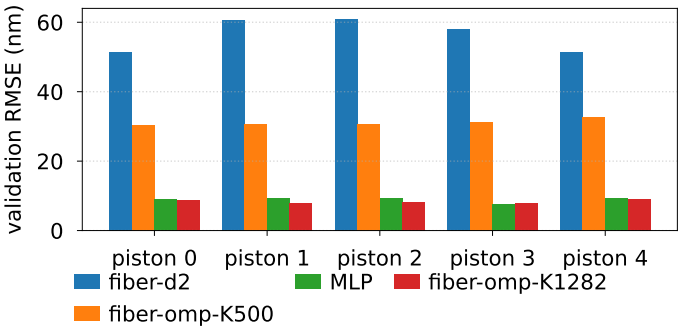




- $19 \to 3 \times 128 \to 5$, Adam, 500 epochs, best validation epoch kept: **9.0 nm**.
- Hyperparameter search (4×256, cosine schedule): $\sim$4.3 nm on the same data; more data helps more than bigger networks (appendix).
- Errors are heavy-tailed: median $\ll$ RMSE.

[old ranges] numbers and plots from the first campaign ($\pm$300 / $\pm$500 nm single, ±5 μm spectral); the rerun at $\pm\lambda/2$ and $\pm8\lambda$ is running and replaces them.

Single wavelength: comparison (19 ports, 800 nm, $\pm\lambda/2$)



reconstructor	RMSE (nm)
no skill	173
linear	103.1
dense quadratic	56.5
OMP cubic, K=500	31.1
MLP	**9.0**
OMP cubic, K=1282	**8.4**

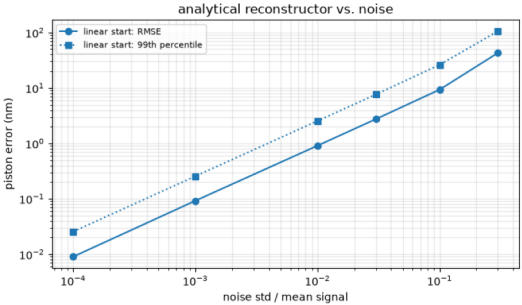
A sparse cubic polynomial matches the MLP with $\sim$6$\times$ fewer parameters.

[old ranges] numbers and plots from the first campaign ($\pm$300 / $\pm$500 nm single, $\pm$5 μm spectral); the rerun at $\pm\lambda/2$ and $\pm 8\lambda$ is running and replaces them.

Physics-based and symbolic reconstructors

- **Closed form**: with petal pistons the signals are exactly quadratic forms in the petal phasors, $s_k = \sum_{j,j'} e^{i(\theta_j - \theta_{j'})} H^{(k)}_{jj'}$: 619 constants for 19 cores and 6 petals, fitted from data.
- Inverted by Gauss-Newton from a few starts; converges to $\ll 1$ nm noiseless.
- **Symbolic regression** (PySR, Operon) finds compact per-piston formulas; useful for insight, less accurate.

PLACEHOLDER: decide how much of the closed-form / symbolic work goes in the main talk vs appendix; add its accuracy-vs-noise numbers.



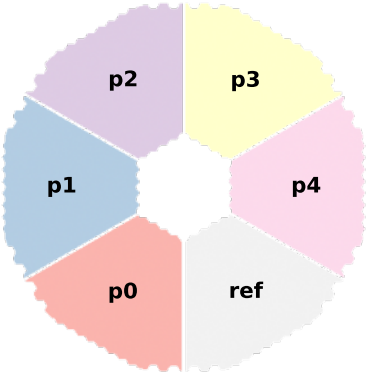
closed-form model: error vs. signal noise (19 ports, 800 nm)

[old ranges] numbers and plots from the first campaign (± 300 / ± 500 nm single, ± 5 μm spectral); the rerun at $\pm\lambda/2$ and $\pm 8\lambda$ is running and replaces them.

ANDES and the petal problem

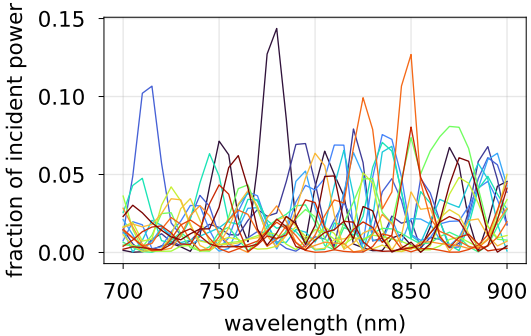
- ELT pupil: the spiders split it into **6 petals**; differential pistons between them ("petal modes") are poorly sensed by the pyramid WFS and can be large (low-wind effect, M4 segment phasing).
- We consider 5 petal pistons (the 6th petal as reference), from the ANDES DM modal basis.
- Target: sense pistons over **many waves** ($\pm 8\lambda$: $\pm 6.4\,\mu$m at 800 nm, $\pm 9.6\,\mu$m at 1200 nm), not only inside $\pm\lambda/2$.

PLACEHOLDER: ANDES-specific requirements: expected petal piston amplitudes and temporal behaviour, wavelength band available for sensing, required accuracy (nm) and photon budget – from the ANDES AO/SCAO team.



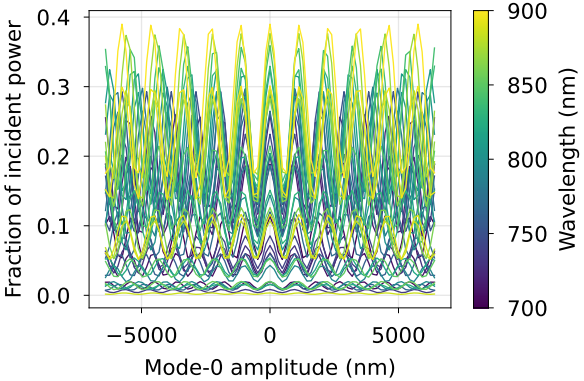
Beyond $\pm\lambda/2$: the dispersed lantern

- At one wavelength a piston is known only modulo λ.
- Feed the N lantern outputs to a spectrograph slit: each core gives a **spectrum**, i.e. the signals at many wavelengths at once.
- Different λ wrap differently $\Rightarrow$ synthetic-wavelength interferometry can resolve the full range.
- We simulate 41 wavelengths (5 nm steps) over the band; the power of each core at each wavelength, as a fraction of the incident power.



one random $\pm 8\lambda$ configuration: the 19 core spectra (19 ports, 700–900 nm)

Spectral response to one petal piston



- Petal 0 ramped over $\pm 8\lambda$ ($\pm 6.4\,\mu$m); the central core's signal at each of the 41 wavelengths.
- Each wavelength is periodic in the piston with its own period λ: the combination is unique over the range.

The same response for the 1200 nm lanterns: appendix.

Three lanterns

	pl19_800 (reference)	**pl7_1200**	**pl19_1200**
ports	19	7 (1 + 6 hexagon)	19
design λ / band	800 / 700–900 nm	1200 / 1100–1300 nm	1200 / 1100–1300 nm
cladding / core / jacket Δn	1.444 / +8.8 / $-5.5\,(10^{-3})$	1.4481 / +5.0 / $-10\,(10^{-3})$	as pl7_1200
multimode NA, radius	0.126, 9.0 μm	0.170, 6.4 μm	0.170, 10.0 μm
V at design λ	8.90	5.69	8.90
guided modes (blue / centre / red)	30 / 23 / 17	10 / 10 / 8	23 / 23 / 17
taper, length	$\times$12, 100 mm	$\times$15, 100 mm	$\times$15, 100 mm
flat-wavefront coupling	34–53%	41–57%	36–50%

- The 1200 nm lanterns share one fibre technology (Ge-doped SMF-like cores, F-doped capillary); **only the port count differs** between them.
- Same PSF sizing rule for all: beam NA / multimode NA as in the reference.

Design choices at 1200 nm

- Multimode end sized to the port count: $\sim N$ (to $N{+}4$) guided modes.
- 7 ports: LP_{31} comes as a pair; V chosen so it stays guided to 1300 nm.
- Output cores: 6.6 μm, single-mode over the band; pitch 60 μm (uncoupled).
- Mode bookkeeping (degenerate pairs, dropped mode) checked by FEM every 25 nm.
- Mesh: $\Delta n_{\mathrm{eff}} \leq 1.6 \times 10^{-5}$ when doubled.

7-port cladding radius (FEM)

r_{clad}	1100	1200	1300 nm
6.2 μm	10	10	6
6.4 μm	10	10	**8**
6.6 μm	12	10	8

guided modes at the multimode end

PLACEHOLDER: design references: Lin et al. 2022/2023 (6- and 19-port lanterns, $V \approx 5.1$ and 9.6); Birks et al. 2015.

Simulation campaign

- Per lantern: 41 characterizations (one per wavelength) $\rightarrow$ two datasets:
  - **single wavelength**: 10 000 random 5-piston configurations, camera signals of the cores at the design λ, $\pm\lambda/2$ ($\pm$400 nm at 800 nm, $\pm$600 nm at 1200 nm);
  - **spectral**: 50 000 configurations, $\pm 8\lambda$ ($\pm$6.4 / $\pm$9.6 μm), core powers at 41 wavelengths ($\lambda \pm 100$ nm).
- 40 000 / 10 000 train / validation; all reconstructors trained on identical data.
- Runs on CINECA Leonardo (Booster): the 41 characterizations and the dataset pieces spread over whole nodes, ≤ 8 nodes at a time; first campaign on a 64-core / 2-GPU server.
- One code base, lantern chosen by a preset name; executed notebooks kept per lantern.

PLACEHOLDER: add photon noise / detector noise and a spectrograph model (the `spectral` package exists; not yet in the reconstruction study).

Two strategies for many wavelengths

A. One model on the whole spectrum

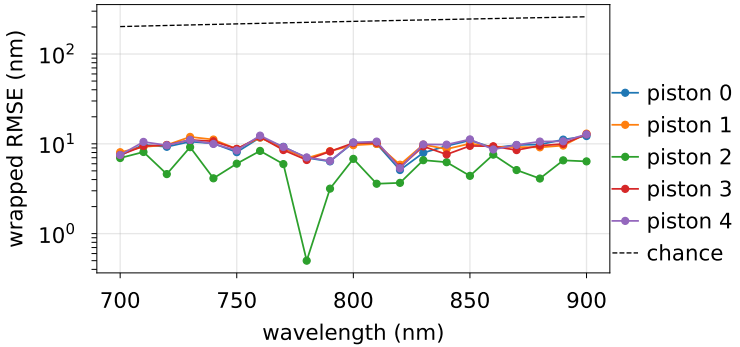
Input: all $41 \times N$ signals; output: the 5 pistons directly. Linear, dense quadratic ridge (wavelength subset), global sparse cubic (OMP), MLP.

B. Per wavelength, then unwrap

At each λ: a small model regresses each piston's *phasor* $(\sin, \cos)(2\pi p/\lambda)$ – smooth, bounded, one fringe. Then a multi-wavelength unwrap (circular maximum likelihood over $\pm 8\lambda$ + robust refinement) fuses the wrapped estimates, as in synthetic-wavelength interferometry.

- B with sparse cubic polynomials (OMP) or small MLPs per wavelength.
- A "wrong fringe": error larger than half the shortest wavelength.

Per-wavelength models (19 ports, 700–900 nm)

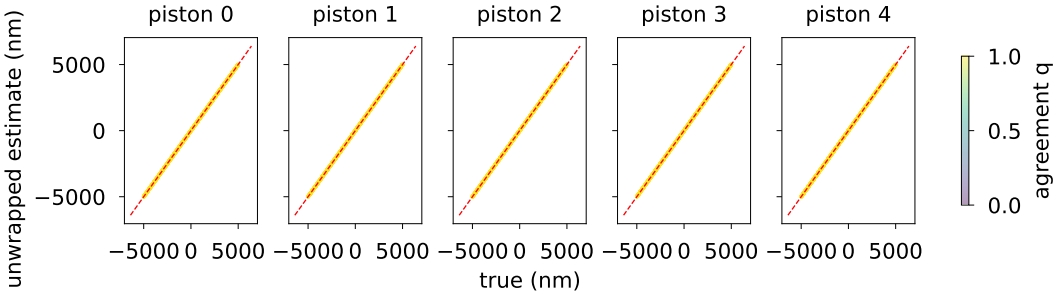


wrapped-piston RMSE of each per-wavelength model (dashed: chance)

- Each per-wavelength model gets the wrapped piston (one fringe) to a few nm.

[old ranges] numbers and plots from the first campaign ($\pm$300 / $\pm$500 nm single, $\pm 5\ \mu$m spectral); the rerun at $\pm\lambda/2$ and $\pm 8\lambda$ is running and replaces them.

The multi-wavelength unwrap (19 ports, 700–900 nm)

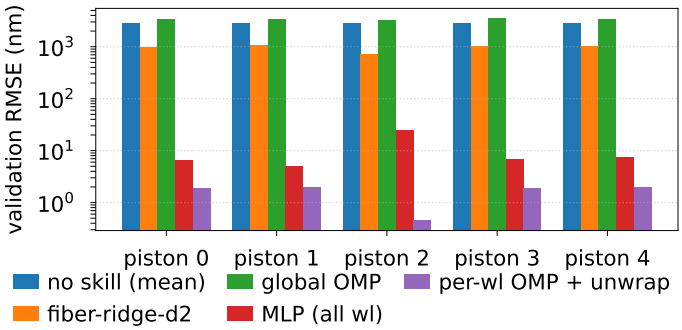


unwrapped estimate vs. truth over the full $\pm 8\lambda$, one panel per piston

- The unwrap fuses the wrapped estimates of the 41 wavelengths and recovers the full range with **no wrong fringes**: **1.64 nm** rms.

[old ranges] numbers and plots from the first campaign (± 300 / ± 500 nm single, ± 5 μm spectral); the rerun at $\pm\lambda/2$ and $\pm 8\lambda$ is running and replaces them.

Comparison, reference lantern (19 ports, 700–900 nm, $\pm 8\lambda$)



approach	RMSE (nm)
no skill	2893
global OMP (3 λ)	3420
linear	2809
dense quadratic	969
MLP, full spectrum	10.25
per-λ OMP + unwrap	**1.64**
per-λ MLP + unwrap	0.44–1.0

last row: hyperparameter search, held-out protocol

[old ranges] numbers and plots from the first campaign (± 300 / ± 500 nm single, ± 5 μm spectral); the rerun at $\pm \lambda/2$ and $\pm 8\lambda$ is running and replaces them.

7 ports at 1200 nm: results

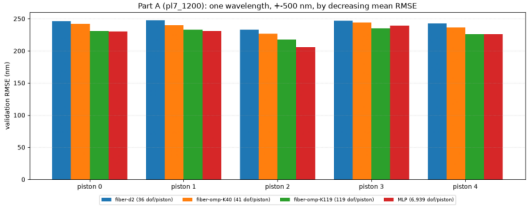
	single λ $\pm\lambda/2$	spectral $\pm 8\lambda$
no skill	288	2893
linear	261.9	2852
dense quadratic	243.2	1488
OMP (cubic / global)	228.5	2249
MLP	226.4	**278.0**
per-λ OMP + unwrap	–	1665

nm rms, validation set. $r_{\mathrm{clad}} = 6.4$ μm; the earlier 6.2 μm design gave 221.8 / 257.8 / 1688 nm for the MLP / full-spectrum MLP / unwrap: the redesign changes little.

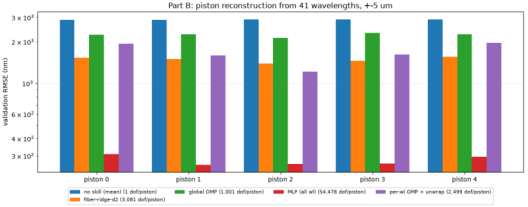
- One wavelength: every reconstructor stops at $\sim$230 nm (no skill 288); the cubic OMP sweep is flat (260 $\rightarrow$ 229 nm from $K = 5$ to 119): the model is not the limit.
- Spectral: the per-wavelength route **fails** – 53% of the estimates on a wrong fringe; only the full-spectrum MLP learns (10$\times$ better than no skill; still improving at its last epoch).
- Flat-wavefront coupling 41–57% over the band.

[old ranges] numbers and plots from the first campaign ($\pm$300 / $\pm$500 nm single, ± 5 μm spectral); the rerun at $\pm\lambda/2$ and $\pm 8\lambda$ is running and replaces them.

7 ports at 1200 nm: per piston



one wavelength (1200 nm), $\pm\lambda/2$



41 wavelengths (1100–1300 nm), $\pm 8\lambda$ (log scale)

Per-wavelength models and their unwrap: appendix.

[old ranges] numbers and plots from the first campaign (± 300 / ± 500 nm single, $\pm 5\ \mu$m spectral); the rerun at $\pm\lambda/2$ and $\pm 8\lambda$ is running and replaces them.

19 ports at 1200 nm: results

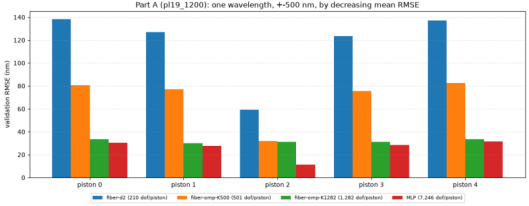
	single λ $\pm\lambda/2$	spectral $\pm 8\lambda$
no skill	288	2893
linear	213.3	2767
dense quadratic	117.1	397
OMP (cubic K=1282 / global)	31.9	1827
MLP	**25.9**	**9.48**
per-λ OMP + unwrap	–	30.0

nm rms, validation set; same fibres, wavelength and data as pl7_1200 – only the port count differs.

- 19 ports restore the performance: **9.5 nm** over the spectral range with the full-spectrum MLP (7 ports: 278 nm; reference lantern: 10.25 nm).
- One wavelength: 26–32 nm. In the first campaign not directly comparable with the reference's 8–9 nm ($\pm 0.375\lambda$ vs $\pm 0.42\lambda$); the rerun uses $\pm\lambda/2$ for both.
- Per-λ OMP + unwrap: the bulk of the estimates is at the nm level (piston 2: 0.65 nm), but $\sim$1 in 1000 lands on a wrong fringe and dominates the RMSE (30 nm).

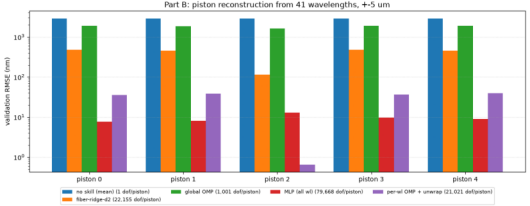
[old ranges] numbers and plots from the first campaign (± 300 / ± 500 nm single, ± 5 μm spectral); the rerun at $\pm\lambda/2$ and $\pm 8\lambda$ is running and replaces them.

19 ports at 1200 nm: per piston



one wavelength (1200 nm), $\pm\lambda/2$

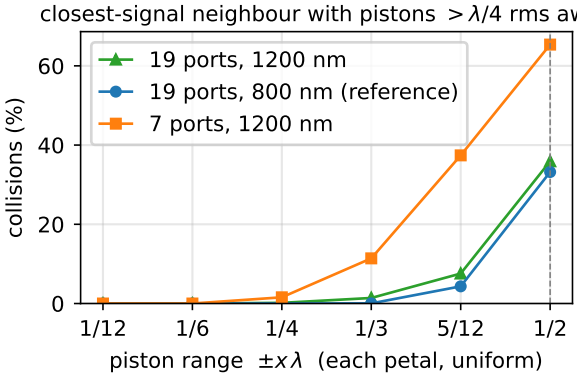
Per-wavelength models and their unwrap: appendix.



41 wavelengths (1100–1300 nm), $\pm 8\lambda$ (log scale)

[old ranges] numbers and plots from the first campaign ($\pm$300 / $\pm$500 nm single, $\pm 5\ \mu$m spectral); the rerun at $\pm\lambda/2$ and $\pm 8\lambda$ is running and replaces them.

Why do 7 ports fail? Not resolution: uniqueness



- Test, no reconstructor involved: 3000 random piston sets per range; for each, the set with the closest signals. A **collision** = their pistons differ by $> \lambda/4$ rms (the signals point to a wrong solution).
- Locally the 7-port is fine (full-rank Jacobian, twin image broken); no collisions up to $\pm\lambda/6$.
- Up to $\pm 5\lambda/12$: **37%** collisions with 7 ports, 7.6% with 19 ports in the same fibres, 4.3% for the reference.
- At the study's $\pm\lambda/2$ (dashed) the wrap adds its own: $+\lambda/2 \equiv -\lambda/2$. **65%** with 7 ports, 36% / 33% with 19 ports.
- About 1/3 of the collisions are twin-image pairs; the rest are other folds.

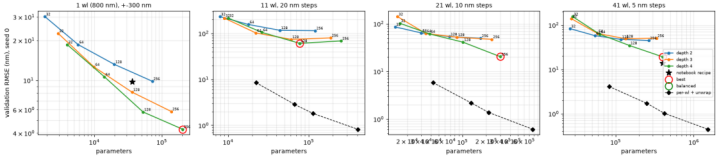
Why do 7 ports fail? Counting dimensions

- The 5 pistons live on a 5-dimensional torus (each wraps with λ).
- An N-port lantern gives $N-1$ independent normalised signals per wavelength.
- A smooth map of a 5-dimensional space is in general one-to-one only into $\geq 2 \cdot 5 + 1 = 11$ dimensions (Whitney); into fewer it folds onto itself, with families of colliding configurations of dimension $2 \cdot 5 - (N-1)$.
- **19 ports: 18 $\geq$ 11** $\Rightarrow$ unique. **7 ports: 6 $<$ 11** $\Rightarrow$ 4-dimensional families of piston sets with identical signals.
- Consistent with the literature: an N-port lantern senses at most $\sim N-1$ modes per frame (Lin et al. 2022, 2023).
- Per-wavelength models see the pistons wrapped over a full $\pm\lambda/2$ $\Rightarrow$ they inherit the ambiguity; a model of the whole spectrum (41×6 numbers) can still partly resolve it.

Confirmed by the 1200 nm pair

Same fibres, same wavelength, only the port count differs: at $\pm 5\lambda/12$ the collisions drop from 37% (7 ports) to 7.6% (19 ports), at $\pm\lambda/2$ from 65% to 36%; the best spectral reconstruction from 278 nm to 9.5 nm **[old ranges]**.

Model size vs accuracy (19 ports, MLP hyperparameter search)



- Full-spectrum MLPs need depth and are hard to train (seed spread).
- Per-wavelength MLPs + unwrap: 0.44–1.0 nm, also with 20 nm steps (11 wavelengths).

PLACEHOLDER: repeat the search for pl7_1200 / pl19_1200 (needs porting `scripts/mlp_hpo.py`).

[old ranges] numbers and plots from the first campaign (± 300 / ± 500 nm single, ± 5 μm spectral); the rerun at $\pm\lambda/2$ and $\pm 8\lambda$ is running and replaces them.

Comparison across lanterns

RMSE (nm)	pl19_800	pl7_1200	pl19_1200
single λ, best (MLP / OMP)	9.0 / 8.4	226.4 / 228.5	25.9 / 31.9
single λ, no skill	173	288	288
single λ, range (rerun)	$\pm\lambda/2$	$\pm\lambda/2$	$\pm\lambda/2$
collisions at $\pm\lambda/2$	33%	65%	36%
collisions at $\pm 5\lambda/12$	4.3%	37%	7.6%
spectral, full-spectrum MLP	10.25	278.0	**9.48**
spectral, per-λ OMP + unwrap	1.64	1665 (53% wrong)	30.0 (0.08% wrong)
spectral, no skill	2893	2893	2893

PLACEHOLDER: robustness: photon noise, read noise, calibration errors, wavelength sampling (20/10/5 nm), lantern manufacturing variations.

[old ranges] numbers and plots from the first campaign ($\pm$300 / $\pm$500 nm single, $\pm 5\ \mu$m spectral); the rerun at $\pm\lambda/2$ and $\pm 8\lambda$ is running and replaces them.

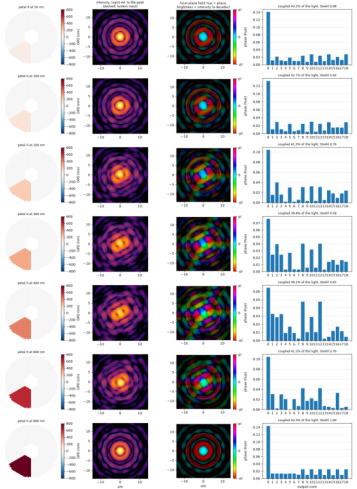
Conclusions (so far)

1. A photonic lantern turns the focal-plane field into N intensities through fixed modal interference; simulated end-to-end with coupled-mode theory (cbeam).
2. At one wavelength, a 19-port lantern supports nonlinear reconstructors at $\sim$8–9 nm over $\pm 0.375\lambda$ **[old ranges]**; sparse cubic polynomials match MLPs. Over the full $\pm\lambda/2$ the wrap at the edges adds unavoidable ambiguous cases.
3. Dispersing the outputs extends the range to many waves: per-wavelength models + unwrap reach $\sim$1–2 nm over $\pm 5\,\mu$m (19 ports) **[old ranges]**.
4. The **port count** matters: with 7 ports the signals are not unique over a large range; only joint use of all wavelengths partly helps (278 nm **[old ranges]**).
5. At 1200 nm with the same fibres, 19 ports bring the spectral error back to $\sim$10 nm (7 ports: 278 nm) **[old ranges]**: the port count, not the wavelength or the fibres, sets the limit.
6. **PLACEHOLDER:** recommendation for ANDES (with the team).

Next steps

- MLP hyperparameter search for pl7_1200 and pl19_1200 (port `scripts/mlp_hpo.py`).
- Noise: photon / read noise through the spectrograph model; required flux.
- Calibration on a real device: how many calibration samples does each approach need?
- Sensitivity to fabrication: random lanterns of the same design.
- Closed loop with the ANDES AO simulation.
- **PLACEHOLDER:** others from the team.

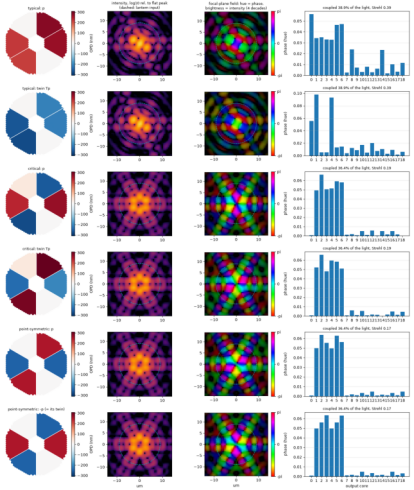
Appendix: more on the lantern response



- Petal 0 from 50 to 800 nm: pupil, focal-plane intensity and phase, power in each output core (19 ports, 800 nm).
- At 800 nm (one wave) the response returns to the flat one.

[old ranges] numbers and plots from the first campaign (± 300 / ± 500 nm single, ± 5 μm spectral); the rerun at $\pm\lambda/2$ and $\pm 8\lambda$ is running and replaces them.

Appendix: twin pairs

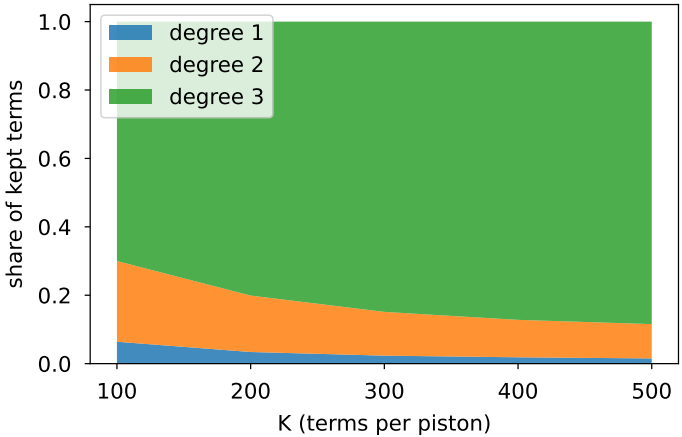


Appendix: PSF size and coupling

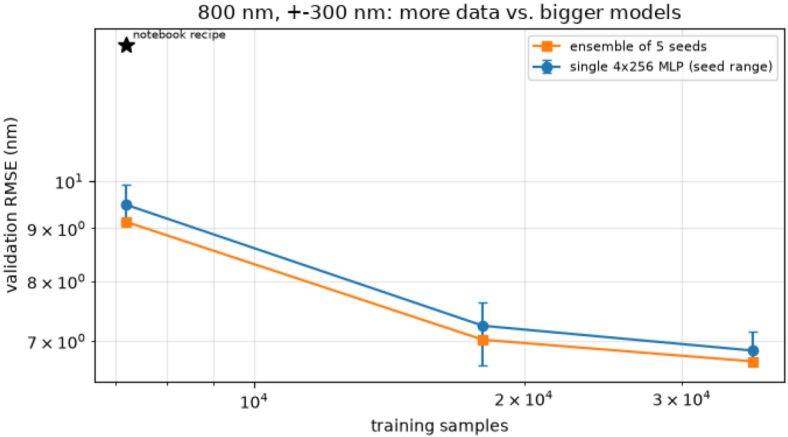
- The PSF is sized so that the Airy first-null radius is $0.30\, r_{\mathrm{clad}}$ for the reference lantern; the 1200 nm lanterns keep the same beam-NA / fibre-NA ratio (fill $= 0.30\, V_{\mathrm{ref}}/V$).
- This overfills the fibre NA by $\sim$1.4$\times$: flat-wavefront coupling $\sim$45%, vs. $\sim$90% for a throughput-optimised focal ratio (Lin et al. 2023).
- Kept for comparability with the reference; its effect on sensing is to be studied.

PLACEHOLDER: scan of reconstruction accuracy and coupling vs. focal ratio.

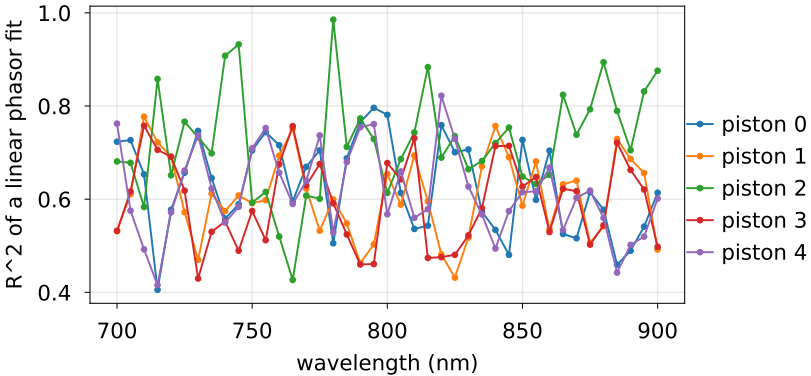
Appendix: OMP – which monomials are kept?



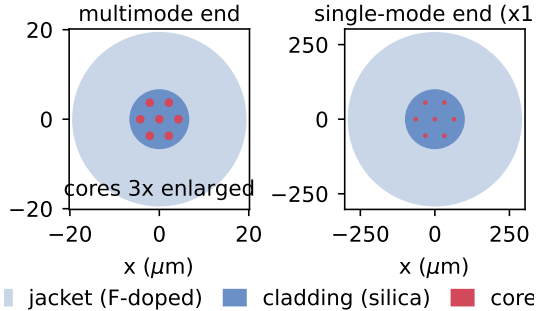
Appendix: more data vs bigger networks (19 ports, 800 nm)

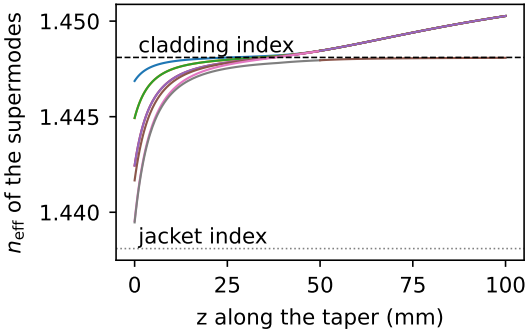


Appendix: linear sensitivity of each wavelength to each piston

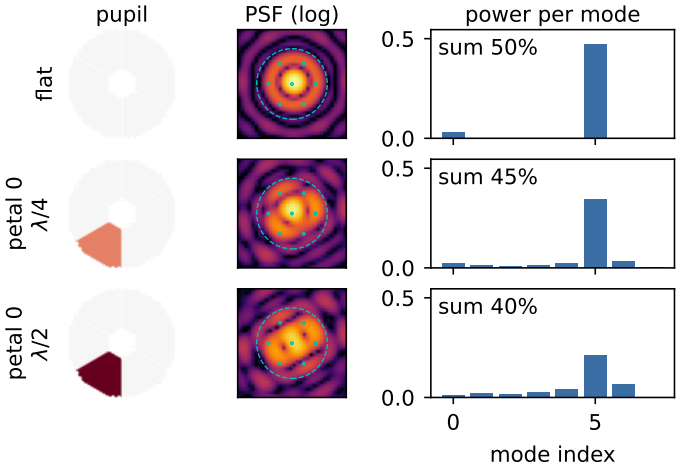


Appendix: 7 ports at 1200 nm in simulation

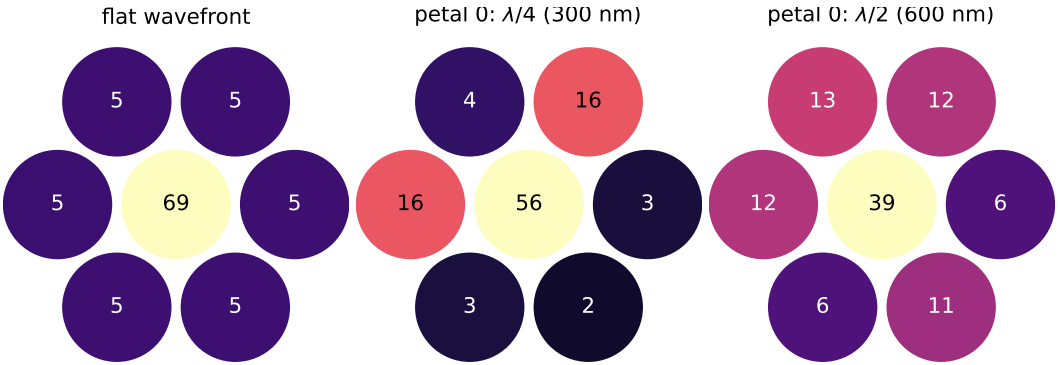




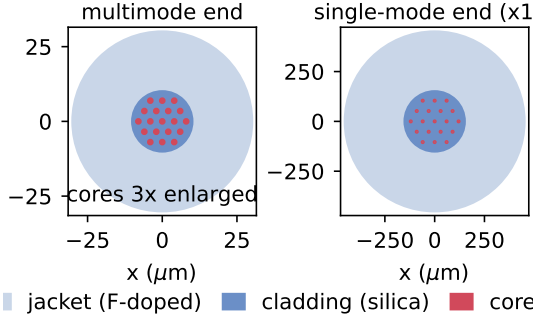
Appendix: one petal piston through the 7-port lantern at 1200 nm

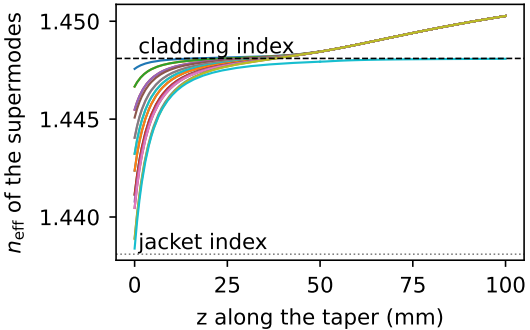


Appendix: 7-port lantern at 1200 nm, output face

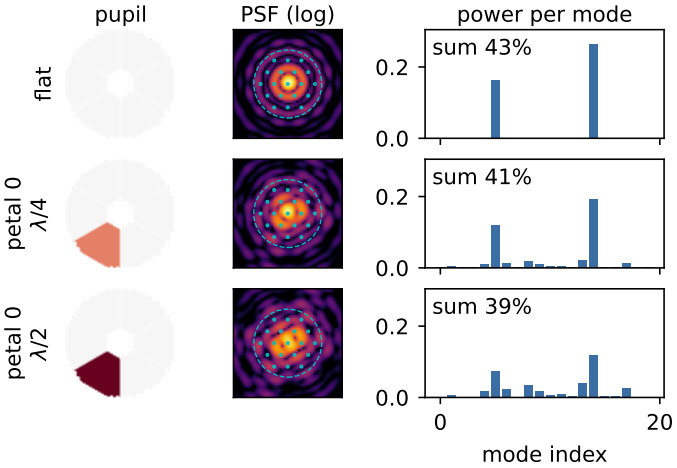


Appendix: 19 ports at 1200 nm in simulation



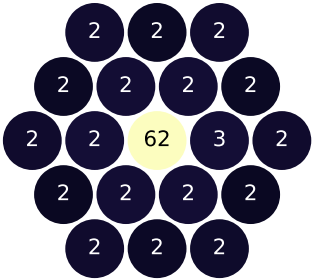


Appendix: one petal piston through the 19-port lantern at 1200 nm

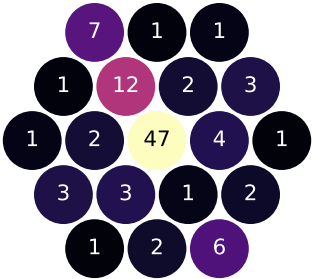


Appendix: 19-port lantern at 1200 nm, output face

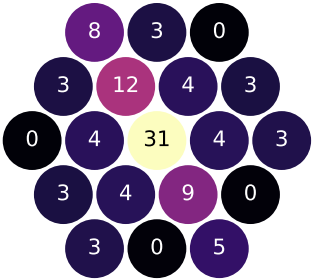
flat wavefront



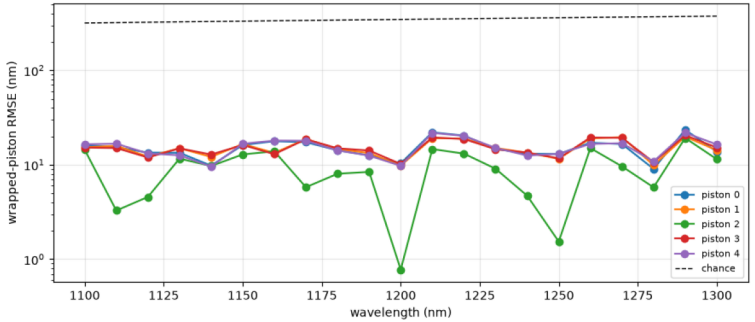
petal 0: $\lambda/4$ (300 nm)



petal 0: $\lambda/2$ (600 nm)



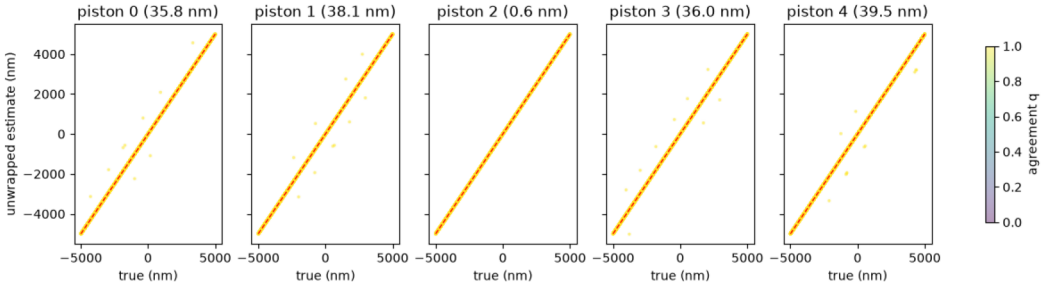
Appendix: 19 ports at 1200 nm, per-wavelength models



per-wavelength models: wrapped-piston error (dashed: chance)

[old ranges] numbers and plots from the first campaign (± 300 / ± 500 nm single, ± 5 μm spectral); the rerun at $\pm\lambda/2$ and $\pm 8\lambda$ is running and replaces them.

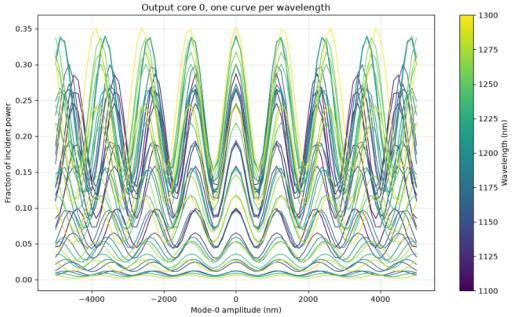
Appendix: 19 ports at 1200 nm, the unwrap



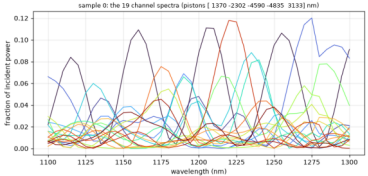
unwrap: a few wrong-fringe outliers per piston

[old ranges] numbers and plots from the first campaign (± 300 / ± 500 nm single, ± 5 μm spectral); the rerun at $\pm\lambda/2$ and $\pm 8\lambda$ is running and replaces them.

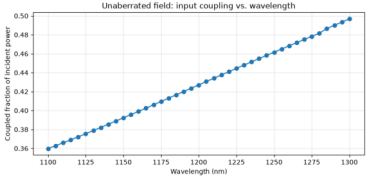
Appendix: 19 ports at 1200 nm, spectral response



central core, one petal ramped over $\pm 8\lambda$ ($\pm 9.6\,\mu$m), one curve per wavelength

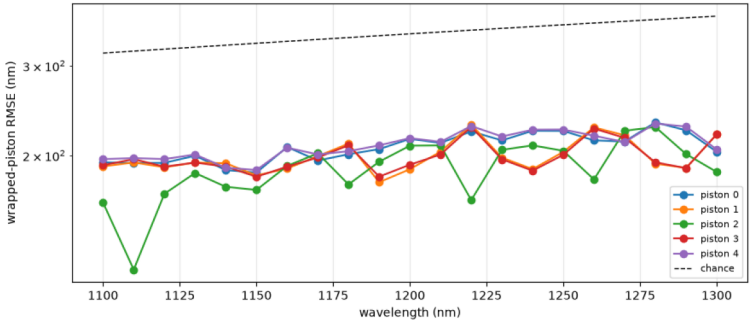


one random configuration: the 19 core spectra



flat-wavefront coupling vs. wavelength

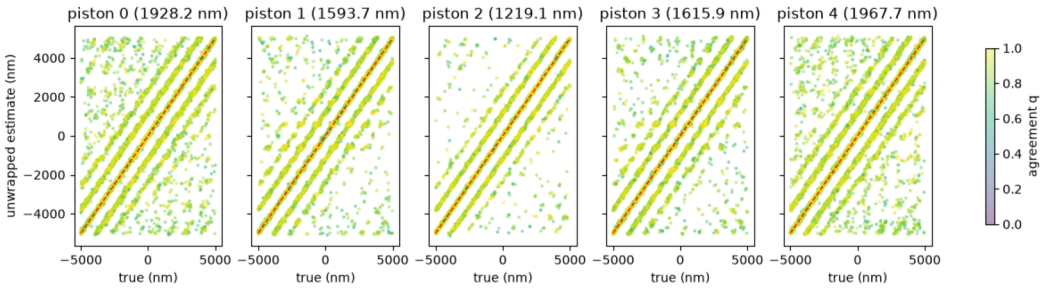
Appendix: 7 ports at 1200 nm, per-wavelength models



per-wavelength models: wrapped-piston error ∼200 nm (dashed: chance)

[old ranges] numbers and plots from the first campaign (± 300 / ± 500 nm single, $\pm 5\ \mu$m spectral); the rerun at $\pm\lambda/2$ and $\pm 8\lambda$ is running and replaces them.

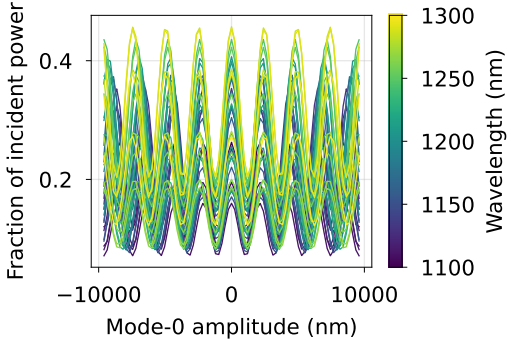
Appendix: 7 ports at 1200 nm, the unwrap



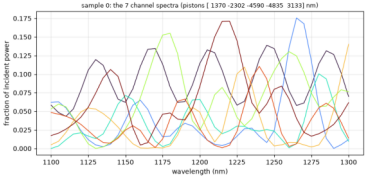
the unwrap then lands on wrong fringes

[old ranges] numbers and plots from the first campaign (± 300 / ± 500 nm single, ± 5 μm spectral); the rerun at $\pm\lambda/2$ and $\pm 8\lambda$ is running and replaces them.

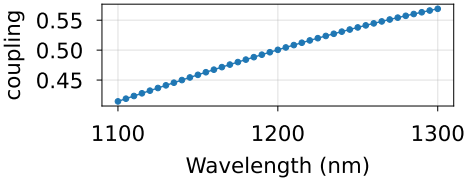
Appendix: 7 ports at 1200 nm, spectral response



central core, one petal ramped over $\pm 8\lambda$ ($\pm 9.6\,\mu$m), one curve per wavelength



one random configuration: the 7 core spectra



flat-wavefront coupling vs. wavelength

Appendix: from Maxwell to the paraxial equation

- Scalar Helmholtz equation (weakly guiding): $\nabla^2 E + k_0^2 n^2 E = 0$.
- Paraxial wave equation for the envelope u:

$$\frac{\partial u}{\partial z} = \frac{i}{2k_0 n_0}\left[\nabla_\perp^2 + k_0^2(n^2 - n_0^2)\right] u$$

- Pupil to focus: $E = \mathcal{F}[Pe^{i\varphi}]$, unitary; aberrations on Zernikes, even or odd under inversion.

Appendix: coupled-mode theory and adiabaticity

- $\frac{da_m}{dz} = i\beta_m a_m - \sum_n \kappa_{mn} a_n, \quad \kappa_{mn} = \langle \xi_m | \partial_z \xi_n \rangle$ (antisymmetric).
- Adiabatic if $|\kappa_{mn}| \ll |\beta_n - \beta_m|$ everywhere; Hellmann-Feynman: $\kappa_{mn} = k_0^2 \langle \xi_m | \partial_z \varepsilon | \xi_n \rangle / (\beta_n^2 - \beta_m^2)$.
- Degenerate pairs aligned by an orthogonal Procrustes step; near the single-mode end the levels merge and the evolution freezes.
- CMT keeps only guided modes (no radiation loss); validate against BPM per design family.

Appendix: wavelength scaling and the sensor as one matrix

- Similarity: $A(\lambda, G) = A(\sigma\lambda, \sigma G)$ if *all* lengths scale, including the taper length; material dispersion breaks it.
- Back-propagate the guided modes to the pupil: $\mathbf{p} = |AM\mathbf{f}|^2 = |C\mathbf{f}|^2$, one precomputed matrix $C(\lambda)$ per wavelength – cheap inside an AO simulation.
- Light outside the N modes $(1 - \sum |u_n|^2)$ is lost to both sensor and science fibres.

Appendix: experimental record

Year	Reference	Result
2025	Lin et al.	On-sky spectral dispersion; improved correction
2023	Lin et al.	Real-time closed loop; petal modes corrected
2020	Norris et al.	19-output lantern; neural-network reconstruction
2018	Corrigan et al.	First laboratory demonstration

PLACEHOLDER: check and complete the references (and add the numerical-characterization papers, Lin et al. 2022 arXiv:2208.10563, 2023 arXiv:2311.01622).